

TEKION INTERVIEW EXPERIENCE

About the Company:

Tekion, founded in 2016 by ex-Tesla CIO Jay Vijayan, is a rapidly growing SaaS company transforming automotive retail with its flagship product, the Automotive Retail Cloud (ARC), the industry's first cloud-native dealer management system.

Read this First:

Tekion mainly focuses on your problem-solving skills and willingness to learn new things. If you don't know something, they don't expect you to be fully updated; instead, they want to see whether you are trainable or not. Their evaluation is centered around DSA problems, core CS fundamentals, and the creative projects you have mentioned in your resume. They are not much concerned about which new technologies, frameworks, or languages you have used in your projects — though having them is, of course, a plus. The company mainly focuses on candidates from the CS branch, though it also hires from IT. Being a CSE student definitely gives you an advantage. (14 people were selected in total)

All Different Rounds:

1. Online Test (Offline)
2. Technical Round – 1 (Offline)
3. Technical Round –2 (Offline)
4. HR Round (Offline)

Online Test:

1. The problem requires finding integer coefficients a and b for each query z from a queries array, such that $ax + by = z$, where x and y are elements from a given array of integers. Additionally, the sum $a + b$ must be less than a given integer m .

If such a and b exist and satisfy the condition, the value $a + b$ is added to a result array; otherwise, -1 is added. This problem can be approached by iterating through all possible pairs of x and y from arr and attempting to solve for a and b for each z in queries. (You may find better test cases in AI do check)

Arr = [2, 3] m = 5 queries = [7, 8, 5]:- Result = [3, 3, 2].

I managed to solve around 8 test cases for this problem. ?

2. You are given two arrays of equal length: process[] and state[]. The process [] array contains alphanumeric process identifiers such as "A1", "B102", etc., while the state [] array contains only "Login" or "Logout", indicating the corresponding action for each process at the same index. A process's activation counts increases by 1 whenever it logs in, and decreases by 1 whenever it logs out, but the count can never go below 0. After processing all the events in order, you are given an integer m, and your task is to determine which processes remain active in the system with an activation count strictly less than m.

```
process = ["B1", "A1", "B1", "B1", "A1"] state = ["Login", "Login", "Login", "Login",  
"Logout"] m = 2  
Result :- ["A1"]
```

I solved this question fully, with all test cases passing.

3. We had 15 MCQs (Java OOPs, DBMS, OS) and the two coding questions above, all on Hacker Rank.

Interview Preparation: -

Honestly, I was not fully prepared for the interview. I thought I wouldn't even get shortlisted since I couldn't solve the first coding problem completely. But to my surprise, the results were announced around 9:30 PM, and I got through. I am strong in DSA and SQL, so I didn't even practice a single problem that day. Instead, I started brushing up DBMS and OS from GeeksforGeeks, going through only the topics I already knew. I also just revised the overview of my projects rather than going in-depth. So, my last-minute preparation was mainly on core subjects like OOPs, DBMS, and OS.

A kind piece of advice: **don't do this!** Try to prepare your core concepts well in advance, ideally during your semester leave, so you won't have to rush at the last moment. My interview process started the next day around 11:50 AM, right after the online test.

Technical round-1: - (1hr)

In the first technical round, the interviewer mainly focused on DSA problems. He quickly went through my resume, asked about the languages and tech stacks I had used in my projects, and then gave me three DSA problems.

Problem 1:

Sort the array in **descending order of frequency** of elements. If two numbers have the same frequency, sort them in **ascending order by value**.

arr = [9, 9, 1, 1, 1, 2, 3, 3] result= [1, 1, 1, 3, 3, 9, 9, 2]

Problem 2:

You are given a matrix (graph) represented with edges. For a given src (source node) and target (destination node), return **true** if there exists a path from src to target, otherwise return **false**.

n = 5 (nodes 0 to 4), edges = [[0,1], [1,2], [2,3], [3,4]], src = 0 target = 4

Ans :- true

Problem 3:

Two sum problem in which we must return each pair which satisfies the sum=k.

I have solved all these problems with the help from the interviewer, and he was satisfied with all my solutions 🤔

Technical round-2: -(1hr 15 mins)

This round was completely focused on **CS fundamentals**. The interviewer went topic by topic and checked how strong I was in the basics.

OOPs:

- What is polymorphism?
- What is static and dynamic binding?
- Explain abstract classes.
- Difference between coupling and cohesion.

DBMS:

- What are transactions?
- Explain ACID properties.
- What is serializability in DBMS?
- Indexing concepts, difference between B-tree and B+ tree.
- What is query optimization and its techniques?
- Sharding, partitioning, and replication.
- Locks in DBMS transactions.

Operating System (OS):

- What is a context switch and why is it important?
- Explain threads and multithreading.
- (Some more OS-based questions I can't recollect).

Computer Networks (CN):

- Difference between IPv4 and IPv6.
- Basics of routing.

SQL:

- Write queries using aggregate functions with joins.
- Implement outer join without using outer join syntax (by using left, right, inner, and cross joins).

DSA:

- Explain Kadane's algorithm.
- Kadane's algorithm in binary tree (maximum path sum in a binary tree).

Projects:

I explained the motivation behind my projects, why I chose them, and some interesting achievements mentioned in my resume. The interviewer was very friendly here and wanted to know more about my thought process and contributions.

☑ I did answer around **four questions incorrectly**, but I solved the **DSA problem much faster** than the interviewer expected, which really impressed him. He appreciated me for that, and it showed that I was very strong in the areas I mentioned at the beginning (DSA and DBMS). This was the **main reason I got shortlisted for the next round**.

HR Round: - (25 mins)

This round was mainly to evaluate how well I fit into **Tekion's culture** and whether I would be a good team player. The interviewer started by asking general questions like *"What is Tekion?"* and *"Why do you want to join us?"* and then moved on to some personal and experiential topics.

She asked about the struggles I had faced so far and how I overcame them. I made sure to highlight both my **technical skills** and **soft skills** during the discussion. She also went through my resume and asked me to explain my projects — for which I gave a brief overview.

At one point, I spoke in detail about my role in a **club position** I had mentioned in my resume. She was interested in knowing about the club and the contributions I had made, which gave me a chance to share some of my leadership and teamwork experiences.

Overall, the experience was very **positive and smooth**. Although I felt nervous initially, my anxiety faded after the first few minutes of conversation because the interviewer was extremely friendly and supportive.

Finally, the overall interview experience was really good. For students with a CGPA of around 7, don't lose hope — focus on studying the CS fundamentals. I am the first student with a 7-pointer CGPA and a history of arrears in CS to get placed within the very first month of the placement drive. So, keep preparing and wait for the right opportunity, your time will come!

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If you have any doubts, please feel free to ask me.